GLOBAL ECONOMICS

10th grade

Learning evidence T2 L2 Decision analysis supplementary material solutions

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C6 Develops decision-making strategies using probabilities and the maximum opportunity criterion.

Problem 1 — Community Bookstore Upgrade

Problem. A community bookstore is deciding how to update its storefront.

- Alternatives (profits in thousands of USD): Expand reading space; Launch a mobile book cart.
- States of nature and probabilities: High foot traffic (0.55); Low foot traffic (0.45).

Payoff table (given):

Alternative	High traffic (0.55)	Low traffic (0.45)
Expand space	48	12
Mobile cart	32	20

Question. Use the maximum opportunity (minimax regret) criterion to choose the best alternative. Then compare the choice to Maximax, Maximin, and Expected Value.

Solution.

(C6) Maximum opportunity criterion (minimax regret). Best payoff in each state:

$$\text{High traffic: } \max\{48, 32\} = 48$$
$$\text{Low traffic: } \max\{12, 20\} = 20$$

Regret table (best payoff − payoff):

Alternative	High traffic	Low traffic	Maximum regret
Expand space	$48 - 48 = 0$	$20 - 12 = 8$	8
Mobile cart	$48 - 32 = 16$	$20 - 20 = 0$	16

Minimax regret choice: *Expand space*, since its maximum regret is 8 (smaller than 16).

Comparison with other criteria.

- **Maximax:** Choose *Expand space* since $\max(48, 12) = 48$ is larger than $\max(32, 20) = 32$.
- **Maximin:** Choose *Mobile cart* since $\min(32, 20) = 20$ is larger than $\min(48, 12) = 12$.
- **Expected Value:** Favors *Expand space* because the higher payoff in the high-traffic state dominates when weighted by the given probabilities.

Problem 2 — Cooperative Packing Line Choice

Problem. A dairy cooperative is choosing a packing line. The profit payoff table below was obtained from estimated revenues and costs under each order size.

- Alternatives: Automated packing line; Manual packing line.
- States of nature (order size): High order (0.30); Medium order (0.45); Low order (0.25).

Payoff table (profit in thousand USD, given):

Alternative	High (0.30)	Medium (0.45)	Low (0.25)
Automated	42	28	8
Manual	30	24	12

Question. Apply the maximum opportunity criterion and then compare with Maximax, Maximin, and Expected Value.

Solution.

(C6) Maximum opportunity criterion (minimax regret). Best payoff in each state:

$$\begin{aligned} \text{High:} &\ \max\{42, 30\} = 42 \\ \text{Medium:} &\ \max\{28, 24\} = 28 \\ \text{Low:} &\ \max\{8, 12\} = 12 \end{aligned}$$

Regret table (best payoff − payoff):

Alternative	High	Medium	Low	Maximum regret
Automated	$42 - 42 = 0$	$28 - 28 = 0$	$12 - 8 = 4$	4
Manual	$42 - 30 = 12$	$28 - 24 = 4$	$12 - 12 = 0$	12

Minimax regret choice: *Automated packing*, since its maximum regret is 4 (smaller than 12).

Comparison with other criteria.

- **Maximax:** Choose *Automated* since its best payoff is 42 (greater than 30).
- **Maximin:** Choose *Manual* since its worst payoff is 12 (greater than 8).
- **Expected Value:** Favors *Automated* because the high and medium outcomes carry more total probability and higher payoffs.

Problem 3 — Craft Export Shipping Plan

Problem. A cooperative that exports handmade crafts must choose a shipping plan. It can ship by air, ship by sea, or delay shipment. Demand and exchange rates combine into four states of nature:

- S1 (high demand + strong currency), $P = 0.20$
- S2 (high demand + weak currency), $P = 0.30$
- S3 (low demand + strong currency), $P = 0.25$
- S4 (low demand + weak currency), $P = 0.25$

Payoff table (profit in thousand USD, given):

Alternative	S1 (0.20)	S2 (0.30)	S3 (0.25)	S4 (0.25)
Air	60	48	20	10
Sea	50	44	26	18
Delay	35	32	28	24

Question. Apply the maximum opportunity criterion and then compare with Maximax, Maximin, and Expected Value.

Solution.

(C6) Maximum opportunity criterion (minimax regret). Best payoff in each state:

$$\begin{aligned} \text{S1:}\ & \max\{60, 50, 35\} = 60 \\ \text{S2:}\ & \max\{48, 44, 32\} = 48 \\ \text{S3:}\ & \max\{20, 26, 28\} = 28 \\ \text{S4:}\ & \max\{10, 18, 24\} = 24 \end{aligned}$$

Regret table (best payoff − payoff):

Alternative	S1	S2	S3	S4	Maximum regret
Air	$60 - 60 = 0$	$48 - 48 = 0$	$28 - 20 = 8$	$24 - 10 = 14$	14
Sea	$60 - 50 = 10$	$48 - 44 = 4$	$28 - 26 = 2$	$24 - 18 = 6$	10
Delay	$60 - 35 = 25$	$48 - 32 = 16$	$28 - 28 = 0$	$24 - 24 = 0$	25

Minimax regret choice: *Ship by sea*, since its maximum regret is 10 (smaller than 14 and 25).

Comparison with other criteria.

- **Maximax:** Choose *Air* since its best payoff is 60 (largest).
- **Maximin:** Choose *Delay* since its worst payoff is 24 (greater than 10 and 18).
- **Expected Value:** Favors *Sea* because it balances payoffs with moderate risk across the four states.

Problem 4 — Campus Vending Options

Problem. A student council is selecting a vending option for campus events. It can choose smart vending kiosks or pop-up snack stalls. Campus traffic can be high ($P = 0.40$), medium ($P = 0.35$), or low ($P = 0.25$). The profit payoff table below was obtained from estimated revenues and costs under each scenario.

Payoff table (profit in thousand USD, given):

Alternative	High (0.40)	Medium (0.35)	Low (0.25)
Kiosks	25	12	−1
Pop-up stalls	23	14	5

Question. Apply the maximum opportunity criterion and then compare with Maximax, Maximin, and Expected Value.

Solution.

(C6) Maximum opportunity criterion (minimax regret). Best payoff in each state:

$$\begin{aligned} \text{High:}\ & \max\{25, 23\} = 25 \\ \text{Medium:}\ & \max\{12, 14\} = 14 \\ \text{Low:}\ & \max\{-1, 5\} = 5 \end{aligned}$$

Regret table (best payoff − payoff):

Alternative	High	Medium	Low	Maximum regret
Kiosks	$25-25=0$	$14-12=2$	$5-(-1)=6$	6
Pop-up stalls	$25-23=2$	$14-14=0$	$5-5=0$	2

Minimax regret choice: *Pop-up stalls*, since its maximum regret is 2 (smaller than 6).

Comparison with other criteria.

- **Maximax:** Choose *Kiosks* since its best payoff is 25 (greater than 23).
- **Maximin:** Choose *Pop-up stalls* since its worst payoff is 5 (greater than −1).
- **Expected Value:** Favors *Pop-up stalls* because the medium and low outcomes have substantial probabilities and higher payoffs than kiosks in those states.

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1.

Problem 1 — Single Project under Uncertainty

Problem. A firm considers launching a project. Profit depends on the state of nature.

	$S_1\,(p)$	$S_2\,(1-p)$
Project	18	−4

Let p be the probability of state S_1, and $1-p$ the probability of state S_2. Determine for which values of p the project is profitable using Expected Value.

Solution. Expected value:

$$EV = 18 \cdot p + (-4) \cdot (1-p)$$

Expand:

$$EV = 18p - 4(1-p) = 18p - 4 + 4p$$

Combine like terms:

$$EV = 22p - 4$$

Profitability condition:

$$EV > 0 \Rightarrow 22p - 4 > 0$$

Solve:

$$22p > 4 \Rightarrow p > \frac{4}{22} = \frac{2}{11}$$

So the project is profitable when $p > \frac{2}{11}$.

2.

Problem 2 — Technology Upgrade

Problem. A company considers upgrading its technology. Profit depends on the state of nature.

	$S_1\,(p)$	$S_2\,(1-p)$
Upgrade	25	-10

Let p be the probability of state S_1. Determine for which values of p the upgrade is profitable.

Solution. Expected value:

$$EV = 25 \cdot p + (-10) \cdot (1-p)$$

Expand:

$$EV = 25p - 10(1-p) = 25p - 10 + 10p$$

Combine like terms:

$$EV = 35p - 10$$

Profitability condition:

$$EV > 0 \Rightarrow 35p - 10 > 0$$

Solve:

$$35p > 10 \Rightarrow p > \frac{10}{35} = \frac{2}{7}$$

So the upgrade is profitable when $p > \frac{2}{7}$.

3.

Problem 3 — Two Investment Alternatives

Problem. Two investment options depend on the state of nature.

	$S_1\,(p)$	$S_2\,(1-p)$
A	22	6
B	16	10

Let p be the probability of state S_1. When is option A better than option B using Expected Value.

Solution. Compute expected value of A:

$$EV(A) = 22 \cdot p + 6 \cdot (1-p)$$

Expand and simplify:

$$EV(A) = 22p + 6 - 6p = (22p - 6p) + 6 = 16p + 6$$

Compute expected value of B:

$$EV(B) = 16 \cdot p + 10 \cdot (1-p)$$

Expand and simplify:

$$EV(B) = 16p + 10 - 10p = (16p - 10p) + 10 = 6p + 10$$

A is better than B when $EV(A) > EV(B)$, equivalently:

$$EV(A) - EV(B) > 0$$

Substitute:

$$(16p + 6) - (6p + 10) > 0$$

Simplify:

$$16p + 6 - 6p - 10 > 0 \Rightarrow 10p - 4 > 0$$

Solve:

$$10p > 4 \Rightarrow p > \frac{4}{10} = \frac{2}{5}$$

So option A is better than option B when $p > \frac{2}{5}$.

4.

Problem 4 — Three Production Plans

Problem. A firm must choose among three production plans.

	$S_1\,(p)$	$S_2\,(1-p)$
A	30	0
B	20	8
C	14	12

Let p be the probability of state S_1. Determine for which values of p each plan is optimal.

Solution. Compute expected value of each plan.

Plan A:

$$EV(A) = 30 \cdot p + 0 \cdot (1 - p) = 30p$$

Plan B:

$$EV(B) = 20 \cdot p + 8 \cdot (1 - p)$$

Expand:

$$EV(B) = 20p + 8 - 8p = 12p + 8$$

Plan C:

$$EV(C) = 14 \cdot p + 12 \cdot (1 - p)$$

Expand:

$$EV(C) = 14p + 12 - 12p = 2p + 12$$

Now compare pairwise using > 0 conditions.

A better than B:

$$EV(A) - EV(B) > 0 \Rightarrow 30p - (12p + 8) > 0$$

Simplify:

$$30p - 12p - 8 > 0 \Rightarrow 18p - 8 > 0 \Rightarrow p > \frac{8}{18} = \frac{4}{9}$$

B better than C:

$$EV(B) - EV(C) > 0 \Rightarrow (12p + 8) - (2p + 12) > 0$$

Simplify:

$$12p + 8 - 2p - 12 > 0 \Rightarrow 10p - 4 > 0 \Rightarrow p > \frac{4}{10} = \frac{2}{5}$$

A better than C:

$$EV(A) - EV(C) > 0 \Rightarrow 30p - (2p + 12) > 0$$

Simplify:

$$30p - 2p - 12 > 0 \Rightarrow 28p - 12 > 0 \Rightarrow p > \frac{12}{28} = \frac{3}{7}$$

Determine optimal plan by intervals:

- If $p < \frac{2}{5}$, then $EV(C) > EV(B)$. Also $EV(C) > EV(A)$ because $EV(A) - EV(C) > 0$ requires $p > \frac{3}{7}$. So for $p < \frac{2}{5}$, plan C is optimal.
- If $\frac{2}{5} < p < \frac{4}{9}$, then $EV(B) > EV(C)$ and $EV(A) < EV(B)$ (since $p < \frac{4}{9}$). So in this interval, plan B is optimal.
- If $p > \frac{4}{9}$, then $EV(A) > EV(B)$ and also $EV(A) > EV(C)$ (since $\frac{4}{9} > \frac{3}{7}$). So for $p > \frac{4}{9}$, plan A is optimal.

5.

Problem 5 — Two Alternatives with Three States

Problem. Two strategies depend on three states of nature.

	$S_1\,(p_1)$	$S_2\,(p_2)$	$S_3\,(1-p_1-p_2)$
A	20	10	-5
B	12	14	0

Let $p_1 = P(S_1)$, $p_2 = P(S_2)$, and $1 - p_1 - p_2 = P(S_3)$. When is strategy A better than strategy B using Expected Value.

Solution. Compute expected value of A:

$$EV(A) = 20p_1 + 10p_2 + (-5)(1 - p_1 - p_2)$$

Expand the last term:

$$EV(A) = 20p_1 + 10p_2 - 5 + 5p_1 + 5p_2$$

Combine like terms:

$$EV(A) = (20p_1 + 5p_1) + (10p_2 + 5p_2) - 5 = 25p_1 + 15p_2 - 5$$

Compute expected value of B:

$$EV(B) = 12p_1 + 14p_2 + 0 \cdot (1 - p_1 - p_2) = 12p_1 + 14p_2$$

A better than B when:

$$EV(A) - EV(B) > 0$$

Substitute:

$$(25p_1 + 15p_2 - 5) - (12p_1 + 14p_2) > 0$$

Simplify:

$$25p_1 + 15p_2 - 5 - 12p_1 - 14p_2 > 0 \Rightarrow 13p_1 + p_2 - 5 > 0$$

So strategy A is better than strategy B when $13p_1 + p_2 - 5 > 0$.

6.

Problem 6 — Policy Choice under Three States

Problem. A government compares two policies under three states of nature.

	$S_1\,(p_1)$	$S_2\,(p_2)$	$S_3\,(1-p_1-p_2)$
A	18	8	-6
B	10	12	2

Let $p_1 = P(S_1)$, $p_2 = P(S_2)$, and $1 - p_1 - p_2 = P(S_3)$. When is policy A better than policy B.

Solution. Compute expected value of A:

$$EV(A) = 18p_1 + 8p_2 + (-6)(1 - p_1 - p_2)$$

Expand:

$$EV(A) = 18p_1 + 8p_2 - 6 + 6p_1 + 6p_2$$

Combine:

$$EV(A) = (18p_1 + 6p_1) + (8p_2 + 6p_2) - 6 = 24p_1 + 14p_2 - 6$$

Compute expected value of B:

$$EV(B) = 10p_1 + 12p_2 + 2(1 - p_1 - p_2)$$

Expand:

$$EV(B) = 10p_1 + 12p_2 + 2 - 2p_1 - 2p_2$$

Combine:

$$EV(B) = (10p_1 - 2p_1) + (12p_2 - 2p_2) + 2 = 8p_1 + 10p_2 + 2$$

A better than B when:

$$EV(A) - EV(B) > 0$$

Substitute:

$$(24p_1 + 14p_2 - 6) - (8p_1 + 10p_2 + 2) > 0$$

Simplify:

$$24p_1 + 14p_2 - 6 - 8p_1 - 10p_2 - 2 > 0 \Rightarrow 16p_1 + 4p_2 - 8 > 0$$

Divide by 4 (positive, inequality direction unchanged):

$$4p_1 + p_2 - 2 > 0$$

So policy A is better than policy B when $4p_1 + p_2 - 2 > 0$.

7.

Problem 7 — New Service Platform Launch

Problem. A company considers launching a digital service. Profit depends on two independent factors: customer adoption and system reliability.

	Stable (q)	Unstable $(1-q)$
High adoption (p)	28	6
Low adoption $(1-p)$	-4	-12

Let p be the probability of high adoption and q the probability of system stability. Find $EV(p,q)$ and determine when launching is profitable.

Solution. By independence, joint probabilities:

$$P(\text{High,Stable}) = pq, \quad P(\text{High,Unstable}) = p(1-q),$$
$$P(\text{Low,Stable}) = (1-p)q, \quad P(\text{Low,Unstable}) = (1-p)(1-q)$$

Expected value:

$$EV = 28(pq) + 6\big(p(1-q)\big) + (-4)\big((1-p)q\big) + (-12)\big((1-p)(1-q)\big)$$

Expand term by term:

$$EV = 28pq + 6p - 6pq - 4q + 4pq - 12(1-p-q+pq)$$

Expand the last product:

$$-12(1-p-q+pq) = -12 + 12p + 12q - 12pq$$

Combine all terms:

$$EV = (28pq - 6pq + 4pq - 12pq) + (6p + 12p) + (-4q + 12q) - 12$$
$$EV = (14pq) + 18p + 8q - 12$$

Profitability condition:

$$EV > 0 \Rightarrow 14pq + 18p + 8q - 12 > 0$$

8.

Problem 8 — Regional Store Opening

Problem. A retailer evaluates opening a new store. Profit depends on demand and competitor response.

	Mild (q)	Aggressive $(1-q)$
Strong demand (p)	35	12
Weak demand $(1-p)$	-5	-20

Let p be the probability of strong demand and q the probability of a mild response. Determine when opening the store is profitable.

Solution. Joint probabilities:

$$P(\text{Strong,Mild}) = pq, \quad P(\text{Strong,Aggressive}) = p(1-q),$$
$$P(\text{Weak,Mild}) = (1-p)q, \quad P(\text{Weak,Aggressive}) = (1-p)(1-q)$$

Expected value:

$$EV = 35(pq) + 12\big(p(1-q)\big) + (-5)\big((1-p)q\big) + (-20)\big((1-p)(1-q)\big)$$

Expand:

$$EV = 35pq + 12p - 12pq - 5q + 5pq - 20(1-p-q+pq)$$

Expand last term:

$$-20(1-p-q+pq) = -20 + 20p + 20q - 20pq$$

Combine:

$$EV = (35pq - 12pq + 5pq - 20pq) + (12p + 20p) + (-5q + 20q) - 20$$

$$EV = 8pq + 32p + 15q - 20$$

Profitability condition:

$$EV > 0 \Rightarrow 8pq + 32p + 15q - 20 > 0$$

9.

Problem 9 — Marketing Strategy Choice

Problem. A firm must choose between two marketing strategies. Outcomes depend on market interest and campaign execution.

State of nature	A	B
High interest (p), Effective (q)	30	22
High interest (p), Ineffective $(1-q)$	10	14
Low interest $(1-p)$, Effective (q)	-6	4
Low interest $(1-p)$, Ineffective $(1-q)$	-14	-8

Let p be the probability of high interest and q the probability of effective execution. When is strategy A better than strategy B.

Solution. Compute EV_A by weighting each payoff:

$$EV_A = 30(pq) + 10\big(p(1-q)\big) + (-6)\big((1-p)q\big) + (-14)\big((1-p)(1-q)\big)$$

Expand:

$$EV_A = 30pq + 10p - 10pq - 6q + 6pq - 14(1 - p - q + pq)$$

Expand last term:

$$-14(1 - p - q + pq) = -14 + 14p + 14q - 14pq$$

Combine:

$$EV_A = (30pq - 10pq + 6pq - 14pq) + (10p + 14p) + (-6q + 14q) - 14$$

$$EV_A = 12pq + 24p + 8q - 14$$

Compute EV_B:

$$EV_B = 22(pq) + 14\big(p(1-q)\big) + 4\big((1-p)q\big) + (-8)\big((1-p)(1-q)\big)$$

Expand:

$$EV_B = 22pq + 14p - 14pq + 4q - 4pq - 8(1 - p - q + pq)$$

Expand last term:

$$-8(1 - p - q + pq) = -8 + 8p + 8q - 8pq$$

Combine:

$$EV_B = (22pq - 14pq - 4pq - 8pq) + (14p + 8p) + (4q + 8q) - 8$$

$$EV_B = -4pq + 22p + 12q - 8$$

Strategy A better than B when:

$$EV_A - EV_B > 0$$

Substitute:

$$(12pq + 24p + 8q - 14) - (-4pq + 22p + 12q - 8) > 0$$

Simplify:

$$12pq + 24p + 8q - 14 + 4pq - 22p - 12q + 8 > 0 \Rightarrow 16pq + 2p - 4q - 6 > 0$$

10.

Problem 10 — Investment Portfolio Selection

Problem. An investor must choose between two portfolios. Outcomes depend on economic growth and interest rates.

State of nature	X	Y
Strong growth (p), Low rates (q)	26	20
Strong growth (p), High rates $(1-q)$	12	16
Weak growth $(1-p)$, Low rates (q)	2	6
Weak growth $(1-p)$, High rates $(1-q)$	-10	-4

Let p be the probability of strong growth and q the probability of low interest rates. When is portfolio X better than portfolio Y.

Solution. Compute EV_X:

$$EV_X = 26(pq) + 12\big(p(1-q)\big) + 2\big((1-p)q\big) + (-10)\big((1-p)(1-q)\big)$$

Expand:

$$EV_X = 26pq + 12p - 12pq + 2q - 2pq - 10(1 - p - q + pq)$$

Expand last term:

$$-10(1 - p - q + pq) = -10 + 10p + 10q - 10pq$$

Combine:

$$EV_X = (26pq - 12pq - 2pq - 10pq) + (12p + 10p) + (2q + 10q) - 10$$

$$EV_X = 2pq + 22p + 12q - 10$$

Compute EV_Y:

$$EV_Y = 20(pq) + 16\big(p(1-q)\big) + 6\big((1-p)q\big) + (-4)\big((1-p)(1-q)\big)$$

Expand:

$$EV_Y = 20pq + 16p - 16pq + 6q - 6pq - 4(1 - p - q + pq)$$

Expand last term:

$$-4(1 - p - q + pq) = -4 + 4p + 4q - 4pq$$

Combine:

$$EV_Y = (20pq - 16pq - 6pq - 4pq) + (16p + 4p) + (6q + 4q) - 4$$

$$EV_Y = -6pq + 20p + 10q - 4$$

Portfolio X better than Y when:

$$EV_X - EV_Y > 0$$

Substitute:

$$(2pq + 22p + 12q - 10) - (-6pq + 20p + 10q - 4) > 0$$

Simplify:

$$2pq + 22p + 12q - 10 + 6pq - 20p - 10q + 4 > 0 \Rightarrow 8pq + 2p + 2q - 6 > 0$$